



EDUCATION

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech	Indian Institute of Technology Kanpur	7.9/10.0
2023	CBSE (XII)	S.K.P. Vidya Vihar, Deoghar	95.2%
2021	CBSE (X)	D.A.V. Public School, Deoghar	95.2%

SCHOLASTIC ACHIEVEMENTS

- Recipient of **Pritam Lal Shakuntala Rawal Memorial Scholarship** for academic excellence (2024, 2025)
- Secured **AIR 6933** in **JEE Advanced 2023** and **AIR 1909** in **JEE Mains 2023** (Top 1%). (2023)

WORK EXPERIENCE

I'mBesideYou Inc.

Software & Data Engineering Intern

May '26 – July '26

- Built a full-stack sales platform (**React 18**, **FastAPI**, **Firestore**) deploying 7 modules; integrated LLM **circuit-breakers**, 400-item batch chunking, and caching for fault-tolerant **Cloud Run** operations.
- Designed **multi-model AI fallback pipeline** (Gemini/Claude/GPT-4o) parsing Zoom transcripts for meeting intelligence, leveraging **OAuth 2.0 webhooks** and async **APScheduler** jobs to eliminate redundant calls.
- Implemented live **5-signal deal health scoring engine** (Recency, Frequency, Progression, Quality, Actions) and stage-adjusted win probability model, updating in real-time across **119 enterprise accounts**.
- Engineered custom export parsers and weighted **mutual-friend strength algorithms** (Gmail, structural, social data) to power a 7-tier warm-intro classification system targeting **C-suite executives**.

TerrAqua UAV Pvt. Ltd.

WebGIS Developer Intern

June '25

- Projected remote sensing imagery onto a live map for real-time **object detection**; integrated **SAM (Segment Anything Model)** for automated image classification and segmentation
- Engineered **WebGIS interface** (React.js + Leaflet) with **GeoJSON/KML** export and polygon area measurement for geospatial data interoperability
- Deployed scalable backend on **AWS** with Cloudinary for geolocation-tagged image storage; built native **Android app** (Capacitor) for offline field data collection

TECHNICAL SKILLS

- Languages:** Python, SQL, C/C++, JavaScript (ES6+), HTML, CSS, Rust (Basic)
- Libraries:** Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, PyTorch, TensorFlow, openpyxl
- Frameworks/Tools:** React.js, Node.js, FastAPI, PostgreSQL, Firebase, Git/GitHub, Docker, AWS

RELEVANT COURSEWORK

Introduction to Machine Learning	Bayesian Modelling and Data Analysis	Computer Vision and Deep Learning
Fundamentals of Computing	Linear Algebra	Ordinary Differential Equations

PROJECTS

Laminar

Operational Intelligence Platform | Inter IIT Tech Meet 14.0

4th/23 IITs

- Built **Laminar**, a real-time fintech operational intelligence platform via **Pathway**; designed low-code workflow builder compiling drag-and-drop pipelines directly to Pathway streaming engine.
- Integrated **Streaming ML** (Tide/Mamba) for predictive SLA risk detection and **Autonomous RCA engine** correlating logs and traces to generate structured incident reports.
- Developed **Context Parser Agent** extracting SLAs from PDFs; implemented four-layer **PII scanning** and output sanitization for financial compliance.

Salient Object Detection in Aerial Images

Course Project | Mentor: Prof. Koteswar Rao

Mar '26 – Apr '26

- Benchmarked **5 DL architectures** for Salient Object Detection on high-res **ORSSD** aerial imagery; designed **RSSODNet** using EfficientNet-B4, ASPP, and cross-scale attention for superior boundary precision.
- Achieved peak recall (**0.8957**) with RSSODNet and highest structural integrity via **BASNet**; evaluated pixel-level accuracy for U-Net, DeepLabV3+, and U2-Net minimizing Mean Absolute Error.

Forecasting using Time Series Analysis

Course Project | CGS698 | Prof. Himanshu Yadav

May '26 – Ongoing

- Analyzed time series data with **Python** (Pandas, NumPy, Matplotlib, Seaborn), covering zero-mean models, linear trends, and stationarity tests.
- Cleaned and preprocessed data by handling missing values and detecting outliers, improving forecast reliability by **18%**.
- Built **ARIMA/SARIMA** models using autoregressive and moving average terms to forecast seasonal and non-stationary data.

ML-PUF Modeling and Delay Recovery

Course Project | CS771 | Prof. Purushottam Kar

May '26 – Ongoing

- Derived a compact **feature map** ($D = 105$) making ML-PUF challenge-response pairs linearly separable; trained **LinearSVC** achieving **99% test accuracy**.
- Recovered 256 intrinsic delay values from 65 observed parameters using a validated **linear reconstruction model**.
- Benchmarked **LinearSVC** and **Logistic Regression** on classification accuracy, training time, and robustness across ML-PUF tasks.

Neuro Mudra Analytics

Mentor: Prof. Rajesh M. Hegde

May '25 – July '25

- Developed **biosignals dataset** from 4 participants (10 mudras); extracted 10+ delta-gamma power and statistical features per **EEG/EMG** channel, enhancing model performance.
- Utilized **LSTM, SVM, Logistic Regression**, and **Random Forest** for mudra classification; achieved **81.82% accuracy** via **Domain Adaptation**, reducing cross-participant domain shift.